Skills Summary

Converting Between Standard Form and Slope-Intercept Form

Use this reference while completing your worksheet or when studying.

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1. Standard → slope-intercept

The two forms: standard $Ax + By = C$ slope-intercept $y = mx + b$

Method: solve for y. Move the x term to the right, then divide *every* term by B:

$$Ax + By = C \implies By = -Ax + C \implies y = -\frac{A}{B}x + \frac{C}{B}$$

So $m = -A/B$ and $b = C/B$ — but derive it each time; the reason to divide every term is that dividing only one changes the line.

Example: Write $2x + y = 8$ in slope-intercept form.
Step 1. The form $y = mx + b$ needs y alone on the left, so undo what is attached to y: here only the $2x$ has to move.
Step 2. Subtract $2x$ from both sides: $y = 8 - 2x$, which reorders to $\boldsymbol{y = -2x + 8}$. Slope -2, y-intercept 8: from $(0, 8)$ the line drops to $(2, 4)$.

Try it: Write $5x + y = 9$ in slope-intercept form.

check: $\boldsymbol{y = -5x + 9}$

2. Slope-intercept → standard

Standard form wants: x and y on the left, a number on the right, **integer** coefficients, and $A > 0$.

Method: (1) clear fractions by multiplying every term by the denominator; (2) move the x term to the left; (3) if A came out negative, multiply the whole equation by -1.

Example: Write $y = 4x - 7$ in standard form.
Step 1. No fractions to clear, so the only work is collecting x and y on the left and keeping A positive.
Step 2. Subtract $4x$: $-4x + y = -7$; multiply by -1 so $A > 0$: $\boldsymbol{4x - y = 7}$. Check at $x = 2$: $y = 1$ and $4(2) - 1 = 7$.

Try it: Write $y = 6x - 2$ in standard form with $A > 0$.

check: $\boldsymbol{6x - y = 2}$

3. Building the equation from a table or two points

Slope from two points: $m = \dfrac{y_2 - y_1}{x_2 - x_1}$ (change in y per 1 unit of x).

Intercept: b is the y-value when $x = 0$. If the table has no $x = 0$ row, substitute one known point into $y = mx + b$ and solve for b.

Example: A tank drains steadily: $(x, y) = (0, 50)$ and $(4, 38)$ litres. Write $y = mx + b$.
Step 1. Two points on a straight line give the slope directly, and one of them already has $x = 0$, so it hands over the intercept as well.
Step 2. $m = \dfrac{38 - 50}{4 - 0} = \dfrac{-12}{4} = -3$ litres per minute, and $b = 50$, so $\boldsymbol{y = -3x + 50}$.

Try it: A candle burns steadily: $(0, 24)$ and $(3, 12)$ centimetres. Write $y = mx + b$.

check: $\boldsymbol{y = -4x + 24}$

4. Converting inside a context (context-conversion)

A context usually arrives in **standard form**, because it counts two things that add to a total: $(\text{cost}_1)x + (\text{cost}_2)y = \text{total}$.

Converting to $y = mx + b$ answers two questions at once: b is how much y you need when $x = 0$, and m is the **trade-off rate** — how much y falls for each extra unit of x.

Example: Tickets cost \$6 (child) and \$3 (adult), and the total is \$60: $6x + 3y = 60$. Convert and read the intercept.
Step 1. The question asks how many adult tickets are needed, so make y the subject — that is exactly what slope-intercept form gives.
Step 2. $3y = 60 - 6x$, divide every term by 3: $y = -2x + 20$, so with no child tickets sold, **20** adult tickets are needed, and each child ticket sold replaces 2 adult tickets.

Try it: $5x + 2y = 50$. Convert to slope-intercept form and state the y-intercept.

check: $\boldsymbol{y = -\frac{5}{2}x + 25}$

(!) Watch out: Divide *every* term by B, not just the x term — $4y = 12 - 3x$ becomes $y = 3 - \frac{3}{4}x$, never $y = 12 - \frac{3}{4}x$. And a negative B flips both signs: $-6x + 2y = 10$ gives $y = 3x + 5$, not $y = -3x + 5$.